Fiscal Year:	FY 2014	**Task Last Updated:**	FY 04/15/2014
PI Name:	Venkateswaran, Kasthuri Ph.D.		
Project Title:	Bacterial, Archaeal, & Fungal Diversity of the ISS		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	(1) Cell & Molecular Biology (2) Microbiology		
Space Biology Cross-Element Discipline:	(1) Reproductive Biology		
Space Biology Special Category:	None		
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City:	Pasadena	**State:**	CA
Zip Code:	91109-8001	**Congressional District:**	27
Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2012 Space Biology NNH12ZTT001N
Start Date:	07/01/2013	**End Date:**	06/30/2017
No. of Post Docs:	1	**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:	1	**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Smith, Jeffrey	**Contact Phone:**	650-604-0880
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Flight Program:	ISS		
Flight Assignment:	NOTE: Title change in May 2015 to "Bacterial, Archaeal, & Fungal Diversity of the ISS--HEPA Filter System" from "Bacterial, Archaeal, & Fungal Diversity of Kibo-JEM Surfaces," per A. Chu/ NASA ARC via NASA JPL Rev. A to Task Order (Ed., 9/24/15) NOTE: Extended to 6/30/2017 per A. Chu/ARC (Ed., 9/17/15)		
Key Personnel Changes/Previous PI:			
COI Name (Institution):	Vaishampayan, Parag Ph.D. (Jet Propulsion Laboratory)		
Grant/Contract No.:	19-12829-27 under Task Order NNN13D111T		
Performance Goal No.:			
Performance Goal Text:			

Task Description:	May 2015 Revision to Task Agreement (Ed. note: information received September 2015). Due to problems obtaining samples from Kibo-JEM, the project has been modified with a new title, “Bacterial, Archaeal, & Fungal Diversity of the ISS--HEPA Filter System.” The revised scope will allow the Jet Propulsion Laboratory (JPL) to undertake a 28-month flight study to examine the microorganisms associated with HEPA filters. The ISS-HEPA project will develop a comprehensive census of the microbial population on the HEPA filter collected from various locations of the International Space Station (ISS), using genetic biosignatures and computational modeling. This microbial census will provide NASA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL team will determine which microbial taxa pose particular threats to crew health. Original Task Description: Space microbiology research working groups in Japan, authorized by the Japan Aerospace Exploration Agency (JAXA), submitted proposals in response to the JAXA Announcement of Opportunity, 2012. The JAXA proposal that is managed by Principal Investigator Prof. Masao Nasu, Osaka University, contained several subgroups, including subgroup 4, headed by Dr. Takashi Yamazaki, JAXA. This subgroup 4 will conduct an investigation on “Analysis of microbial dynamics in the space habitation system by using new molecular microbial ecological techniques and also monitoring microbial contamination from air and surfaces of the ‘Kibo’ Japanese Experiment Module (Kibo-JEM).” The JAXA proposal submitted by subgroup 4 identifies the Jet Propulsion Laboratory (JPL) team headed by Dr. Kasthuri Venkateswaran as a collaborator to jointly explore the investigation. The Bacterial, Archaeal, and Fungal Diversity of Kibo-JEM Surfaces project (JPL-JAXA project) will undertake a three-year flight study to examine the changes in the microorganism population on the Kibo-JEM surfaces. The JPL-JAXA project proposes to develop a comprehensive census of the microbial population on the Kibo-JEM surfaces, using genetic biosignatures and computational modeling. This census will provide NASA and JAXA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL-JAXA team will determine which microbial taxa pose particular threats to crew health. Specific JPLJAXA project goals will enable us to resolve applicable Human Research Program (HRP) integrated research plan (IRP) risks: (1) determine if changes in Kibo-JEM microbiome are occurring during a spaceflight mission and if so, (2) determine whether any changes noted would create a greater health risk for the crew. There are three aims proposed in this JPL-JAXA project and study of the Kibo-JEM microbiome. Through optimizations and comparative evaluations of methods designed to collect, process, analyze, and archive biomolecules sampled, the JPL-JAXA team will increase downstream sensitivity of laboratory assays as well as enhance the current understanding of the population dynamics of the Kibo-JEM module (Aim 1). The JPL-JAXA team will adapt molecular technologies currently in use with the National Institute of Health (NIH) – Human Microbiome Program (HMP) Roadmap—such as rRNA gene-variable region tag sequencing, whole genome shotgun sequencing (WGS), and DNA microarrays, as well as cloning and sequencing of 16S ribosomal DNA (rDNA)—to comprehensively elucidate the microbial diversity profiles associated with the Kibo-JEM surfaces (Aim 2). The final project deliverable to NASA will be the Kibo-JEM database, a compilation of all of the genomic sequences and genetic information of microbes encountered on and within the Kibo-JEM habitat (Aim 3). Computational modeling of the Kibo-JEM microbial population dynamics inferred from the database will then be carried out to predict the complete spectrum of organisms present. The JPL-JAXA project will compare microbial diversity, microbial characteristics, and functional capabilities of the microorganisms associated with the Kibo-JEM. The resulting database will enable HRP to evaluate the effect of spaceflight radiation and microgravity on the Kibo-JEM microbiome. The JPL-JAXA project in combination with results from studies of crew susceptibility will be used to determine the risk of microbiologically induced adverse health effects during a spaceflight mission. This assessment will allow HRP to determine if current operational and engineering controls used to mitigate microbiological risks are adequate or whether additional countermeasures should be developed.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	The JPL-JAXA project proposes to develop a comprehensive census of the microbial population on the Kibo-JEM surfaces, using genetic biosignatures and computational modeling. This microbial census will provide NASA and JAXA with the ability to accurately and confidently assess the status of microbes associated with closed habitation and crew health maintenance. In addition to overall microbial profiles, the JPL-JAXA team will determine which microbial taxa pose particular threats to crew health.
Task Progress:	Maintaining safety of astronauts during long-term habitation in the International Space Station (ISS) is an essential undertaking that must be attained for sustained human presence in space. Of great concern is the ubiquitous presence of microorganisms, which have the potential to not only affect crew health but also have adverse effects on crew systems. Increased microbial burden on ISS could lead to infection or disease outbreaks in the closed habitat. The threat of such an outbreak is especially hazardous due to the limited treatment and patient quarantine options. To date, monitoring the microbial diversity of the ISS was performed utilizing culture-based methods for detailed molecular characterization. The microbial diversity of three distinct closed habitats such as the ISS, Inflatable Lunar-Mars Habitat (ILMH) at University of North Dakota, and a NASA spacecraft assembly facility (SAF) was characterized using the state-of-the-art molecular techniques. Particles accumulated in the vacuum cleaner bags (ISS and SAF) and the floor samples of ILMH were aseptically sampled and analyzed. Cultivation techniques augmented with rapid molecular techniques (qPCR and ATP-assays) were used to measure total, viable, and cultivable microbial burden. Propidium monoazide (PMA) was used to estimate viable microbial burden. In addition, next-generation sequencing technologies were used to elucidate the full spectrum of bacterial, archaeal, and fungal diversity. Microbial burden present in ISS and ILMH are higher than observed in SAF cleanroom environments. Comparatively the ILMH surface samples accumulated more biomass compared to the ISS. Similar trend was observed for the microbial diversity abundance. Archaeal presence was either absent in many samples tested or present at very minimum when compared to fungal and bacterial diversity. The approach adapted during this study was the first ever-comprehensive molecular effort to assess the viability of microbial cells associated with space-related enclosed habitat and to correlate differential viability with phylogenetic affiliation. The results of this study also improved our understanding of background microbial contamination, thus facilitating the development of biosensors to monitor closed habitats like ISS and future manned missions.

Bibliography Type:	Description: (Last Updated: 03/31/2026)
Abstracts for Journals and Proceedings	Cisneros J, Vaishampayan P, de Leon P, Perry J, Venkateswaran K. "Comprehensive Microbial Diversity Characterization of the International Space Station and Analogous Closed Habitats on Earth." Presented at the 114th General Meeting of the American Society for Microbiology, Boston, MA, May 17-20, 2014. ISS - Microbial Observatory. 114th General Meeting of the American Society for Microbiology, Boston, MA, May 17-20, 2014. , May-2014